Pedestrian-Vehicle Interaction Simulation

Team 28

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https://github.com/Austin-Jiang-xlk/CSE-6730-Project

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Abstract

Accurately modeling pedestrian movement remains a primary challenge for autonomous vehicle systems due to the complex nature of human behavior. This behavior is dependent on the context, and traditional physics-based models, such as the Social Force Model, struggle to adapt to new situations. We propose a hybrid multi-modal simulation framework which integrates physics-based dynamics with graph-based learning and adjoint-based optimization. The framework extends the Social Force Model to model specific pedestrian behavioral modes which parameterize motion dynamics. We can then use a Graph Neural Network to infer context and classify behavior modes before employing the Discrete Adjoint-State Method for optimization.

This approach decouples higher-level behavior recognition from lower-level physical dynamics, which we hypothesize will result in improved realism and the ability to generalize across various traffic scenarios. This architecture provides a transparent tool for analyzing pedestrian behavior, which will allow users to make more informed decisions in comparison to preexisting black box solutions.

1. Project description

The system studied in this project is a modular simulation framework for pedestrian - vehicle interaction. The core of this system is a hybrid modeling framework that combines behavior recognition, physical simulation and parameter calibration. We currently have the progress of our research on Github. The backbone of our system is provided by gsfm.py, which uses the Game-theoretic Social Force Model (GSFM) as the underlying dynamics engine. forward simulation is conducted on the movement of pedestrian in mixed traffic scene. This framework is designed for various pedestrian - vehicle trajectory datasets. Currently, it supports CSV data such as DUT/CITR, as well as the larger-scale Shifts dataset, and uses the.npz format for the latter to improve computational efficiency. From the perspective of system functionality, the GSFM part is responsible for deducing the future trajectory of the pedestrian based on the agent state in the scene. It is a forward simulation model for pedestrian dynamics, explicitly including pedestrian-pedestrian and pedestrian-vehicle interactions And anisotropic sensitivity was taken into consideration. This system first assumes that the movement of the pedestrian is driven by a set of interpretable interaction forces, and then uses these forces to update the movement state. This modeling approach has a very direct advantage, that is, the trajectory output by the system can correspond to the specific interaction mechanism, and therefore is more suitable for behavior analysis and safety verification in the scenario of connected autonomous vehicle.

On this basis, our system further introduces the pseudo-labeling mechanism to label the interaction style of the pedestrian with "behavior mode". We have begun to define four types of patterns: aggressive, regular, cautious and following. "aggressive" refers to the situation where a pedestrian is at a disadvantage initially but does not choose to give way, and instead passes first

in the end. “cautious” refers to the situation where the pedestrian originally had the upper hand but significantly slowed down during the interaction and gave the right of way to the vehicle; "following" corresponds to the "crowd following" behavior. The rest of the cases are classified as regular. Therefore, from the perspective of the research object, our system can be regarded as a layered hybrid system. The first layer is the data layer, which is responsible for reading data in different formats and extracting the pedestrian - vehicle conflict scene. The second layer is the interaction understanding layer, which is responsible for generating pseudo labels based on TTC evolution and local motion information, or further generating mode probabilities by classifier. The third layer is the dynamics layer, where GSFM or multimodal GSFM generates future trajectories based on the parameter sets of different modes. Finally, the system also has visualization and animation-related modules for checking the differences between the simulated trajectory and the real interaction process.

2. Literature review

2.1 Background

In the transportation system for connected autonomous vehicles, vehicles need to predict and respond to pedestrian trajectories in a very short time. The problem lies in that the behavior of pedestrians in reality is very difficult to predict. With the rapid deployment of connected and autonomous vehicles (CAVs) in in various traffic contexts, it is anticipated that interactions between human road users (e.g., pedestrians, cyclists, and human drivers) and CAVs will become increasingly frequent in shared spaces, where the explicit traffic rules are absent. (Golchoubian et al., 2023) This trend of ambiguity introduced the importance of understanding pedestrian-vehicle and pedestrian mutual interactions. Pedestrians may exhibit actions such as giving way, tentatively crossing, and accelerating overtake in different traffic contexts. Constructing predictive models that are capable of realistically reproducing such interactions with interpretability can provide decision-making support for autonomous driving systems.

2.2 State of the Art

The existing rule-based pedestrian simulation mainline is represented by the Social Force Model, which generates trajectories using physical terms such as attractive and repulsive forces (Helbing & Molnar, 1995). Early methods, such as social force models (SFM) (Helbing & Molnar, 1995), provided an interpretable, rule-based description of pedestrian interactions through attraction and repulsion, but their reliance on fixed parameters limited their ability to characterize context-dependent behavior in complex pedestrian-vehicle environments. However, such models usually describe the majority of the population with a set of fixed parameters. Therefore, it is difficult for this model to express mode switching in various interaction processes, and the trajectories generated by purely rule-based physical models lack diversity and adaptation. In response to these limitations, in recent years, with the continuous advancement of technology, some scholars have attempted to extend SFM to scenarios involving pedestrians, autonomous vehicles, or

intersections (Yang et al., 2020; Rashid et al., 2024; Shrivas & Kumar, 2024). And introduce the influence of items or interaction mechanisms of the vehicle (Yang et al., 2020; Rashid et al., 2024; Shrivas & Kumar, 2024). At the same time, there are also dedicated trajectory datasets for collecting pedestrian-vehicle interactions for evaluation and calibration (Yang et al., 2019).

To overcome these limitations of rule-based model, data-driven approaches, including recurrent neural networks (Korbmacher et al., 2022), graph neural networks (GNNs) (Sang et al., 2025), and Transformer-based models [5], have been widely adopted to capture rich spatiotemporal interactions between agents. While these methods have achieved good predictive performance, they typically operate as black-box models, lacking explicit physical consistency, which is incompatible to give interference clues to understand the mechanism of pedestrian but also may reduce the trust to the reliability of real-world applications. Furthermore, past models introduced social context into pedestrian modeling to improve accuracy, but examples of such approaches do not explicitly model different behaviors (S. Pellegrini et al., 2009). While these models introduced social context into pedestrian modeling to improve accuracy, examples of such approaches do not explicitly model different behaviors. Other discrete models have clustered pedestrians together in order to predict future motion patterns (Bennewitz et al., 2005). In addition to adding social context to modeling, there has also been work in adding other types of contexts. It is not only other people which effect pedestrian movement, but also other physical objects (curbs, traffic, etc.). Some researchers have earlier included the static physical information as external constraints in simulations (although without force-based dynamics) (Sang et al., 2025). Some researchers have included this information in simulations (although without force-based dynamics) (Kooij et al., 2014). Trajectron++ is another model which uses GNNs to perform trajectory prediction. This model incorporates different types of data beyond just past trajectories to more accurately predict future ones, but it still functions as a black box system (Salzmann et al., 2020).

2.3 Discussion

The review of existing literature reveals a trade-off between predictive flexibility and kinetic rigor. These limitations have inspired the planned design of the proposed hybrid modeling framework. While end-to-end GNNs achieve high accuracy, mapping the graph information onto future position directly, they lead to overfitting to the geometric topology of specific space layouts. A pure GNN trained on intersection data may fail to generalize a roundabout because it learns specific trajectory coordinates rather than agent's mutual interaction principles. This argument inspired us to leverage GNNs exclusively for high-level behavioral mode recognition while delegating the actual trajectory evolution to a Social Force Model (SFM). By extracting underlying mechanical parameters that are scene-agnostic rather than simple coordinate sequences, we can ensure that physical rules remain valid across diverse spatial topologies.

Furthermore, the "black-box" nature of pure neural networks lacks interpretability. In a simulation where a collision occurs, diagnosing the failure within millions of neural parameters

is nearly impossible, posing a risk to the reliability of safety-critical applications. The hybrid framework resolves this by establishing clear accountability through modular design. Errors can be precisely traced. A misclassification of a pedestrian's behavioral mode points to a failure in the GNN perception layer, while a physically unrealistic trajectory in the right recognized mode indicates an issue with the parameters calibrated. By decoupling intention recognition from motion dynamics, the proposed synthesis ensures that simulation outcomes are not only data-driven but also physically compliant and industrially verifiable.

3. Conceptual model

In our hybrid multi-modal simulator for modeling pedestrian–vehicle interactions in complex and highly dynamic environments, the framework integrates physics-based modeling, graph-based learning, and adjoint-based optimization into a unified architecture that preserves interpretability while enabling adaptive, context-aware behavior.

At its core, the simulator is built upon the Social Force Model (SFM), a well-established physics-inspired model that describes pedestrian motion through a system of differential equations governed by interaction forces. In the traditional SFM, agent dynamics are determined by fixed parameters that encode attraction, repulsion, and goal-oriented movement. While this formulation provides strong physical interpretability, it assumes static behavioral rules and therefore cannot capture the context-dependent and multi-modal nature of real human behavior.

To address this limitation, the proposed framework extends the SFM into a multi-modal physical model, in which the governing equations are parameterized by discrete behavioral modes. Each mode corresponds to a distinct interaction regime—such as cooperative navigation, competitive crossing, or group-following behavior—and is represented by a specific set of SFM parameters. This formulation transforms the system into a hybrid dynamical model, where continuous motion is governed by physical laws while discrete transitions between behavioral modes enable adaptive responses to environmental context.

The selection of behavioral modes is achieved through a graph-based perception layer implemented using a Graph Neural Network (GNN). In this representation, the environment is modeled as a dynamic graph in which agents (pedestrians and vehicles) are treated as nodes and their spatial interactions are encoded as edges. The GNN processes this structured representation to infer the current interaction context and classify the corresponding behavioral mode. This approach enables the model to capture complex, multi-agent dependencies without relying on manually defined heuristics, thereby enhancing both scalability and generalization.

To ensure numerical rigor and avoid reliance on black-box training, the framework employs the Discrete Adjoint-State Method for parameter calibration. This method computes gradients efficiently by solving an associated adjoint system backward in time, allowing for precise optimization of model parameters with respect to trajectory-based loss functions. By applying

adjoint-based optimization to each behavioral mode, the framework maintains consistency with physical principles while achieving high-fidelity alignment with empirical data.

Finally, the overall system can be interpreted as a physics-constrained learning architecture, in which machine learning is used exclusively for high-level mode selection, while low-level dynamics remain governed by interpretable physical equations. This separation of roles ensures that the model retains transparency and theoretical grounding, while still benefiting from the adaptability of data-driven methods.

In summary, the proposed conceptual framework integrates (1) a physics-based interaction model, (2) a multi-modal behavioral representation, (3) a graph-based perception mechanism, and (4) an adjoint-based optimization strategy. This combination enables the simulation of realistic, context-dependent pedestrian behavior in a manner that is both computationally efficient and scientifically interpretable, making it well-suited for applications in connected autonomous vehicle systems and intelligent transportation modeling.

3.1 Mathematical Model of GNN + Transformer + MM-GSFM platform

We consider a multi-agent system consisting of N pedestrians. The state of each agent i at time t is defined as:

$$x_i(t) = [p_i(t), v_i(t)]$$

Where $p_i(t)$ denotes position and $v_i(t)$ denotes velocity.

The objective is to model the temporal evolution of all agents:

$$x_i(t) \rightarrow x_i(t + \Delta t)$$

We formulate the dynamics as a combination of physics-based and learned components:

$$dv_i/dt = F_i^{(GSFM)} + F_i^{(GNN)} + F_i^{(Transformer)}$$

$$dp_i/dt = v_i$$

Where $F_i^{(GSFM)}$is the physics-based force from the Multi-Modal Generalized Social Force Model, $F_i^{(GNN)}$ captures spatial interactions via graph neural networks and $F_i^{(Transformer)}$ captures temporal and global dependencies.

To capture heterogeneous pedestrian behaviors, we introduce a multi-modal formulation:

$$F_i^{(GSFM)} = \Sigma_k \pi_i^{(k)} F_i^{(k)}, \Sigma_k \pi_i^{(k)} = 1$$

Where $\pi_i^{(k)}$ represents the probability of behavior mode k. Each mode follows the classical social force decomposition:

$$F_i^{(k)} = F_i^{(goal)} + \Sigma_j F_i j^{(rep)} + F_i^{(env)}$$

The goal force is defined as:

$$F_i^{(goal)} = (v_i^0 e_i - v_i)/\tau_i$$

where is v_i^0desired speed, e_i is desired direction, and τ_i is relaxation time.

At each time step, agents are represented as nodes in a graph G = (V, E), where edges encode proximity-based interactions. Node features are initialized as:

$$h_i^{(0)} = \phi(x_i)$$

Message passing is performed as:

$$h_i^{(l+1)} = \sigma\left(\Sigma_j \alpha_i j W h_j\right)$$

Where α_i denotes attention weights between agents. The resulting interaction force is:

$$F_i^{(GNN)} = f_\theta(h_i)$$

To capture long-range temporal dependencies, we encode trajectory history:

$$H_i = [x_i(t-T), \dots, x_i(t)]$$

A transformer encoder is applied:

$$Attention(Q, K, V) = soft\max\left(QK^T/sqrt(d)\right)V$$

The temporal force contribution is:

$$F_i^{(Transformer)} = g_\varphi(H_i)$$

The system is simulated in discrete time as:

$$v_i^{(t+1)} = v_i^t + \Delta t\left(F_i^{(GSFM)} + F_i^{(GNN)} + F_i^{(Transformer)}\right)$$

$$p_i^{(t+1)} = p_i^t + \Delta t v_i^{(t+1)}$$

This model combines physics-based GSFM, spatial interaction via GNN, and temporal reasoning via Transformer to achieve realistic and robust pedestrian trajectory modeling.

3.2 Platform(s) of development

The platform is structured into four primary layers: a data layer, a learning layer, a physics simulation layer, and an optimization layer. These components are sequentially coupled to form a closed-loop pipeline:

$$Data \rightarrow GraphRepresentation \rightarrow GNN \rightarrow Simulation \rightarrow Optimization.$$

The data layer provides preprocessed trajectory datasets, including DUT, CITR, and Shifts, which are formatted into time-indexed sequences suitable for graph construction. At each time step t, the scene is represented as a graph $\mathcal{G}_t = (\nu_t, \varepsilon_t)$, enabling structured representation of multi-agent interactions.

The overall system operates in a closed-loop training framework:

$$Dataset \rightarrow GNN \rightarrow Simulation \rightarrow Error \rightarrow AdjointOptimization \rightarrow ParameterUpdate.$$

This iterative process ensures that both the mode classification and physical parameters are progressively refined to match observed pedestrian behavior.

The platform is implemented primarily in Python, leveraging its scientific computing ecosystem. Numerical operations are handled using standard libraries such as NumPy and SciPy, while the learning components are implemented using a deep learning framework such as PyTorch. This choice enables efficient tensor computation, automatic differentiation, and scalable model training. The learning layer consists of a Graph Neural Network (GNN) responsible for interaction mode classification. The GNN processes the graph representation Gt and produces a probability distribution over behavioral mode:

$$\hat{\mathbf{y}}_t = \mathrm{softmax}(\mathbf{W}_o \mathbf{h}_{\mathcal{G}} + \mathbf{b}_o),$$

The simulation layer is based on the Social Force Model (SFM), which governs agent motion through a system of differential equations:

$$m\frac{d\mathbf{v}_i}{dt} = \mathbf{F}_{goal} + \mathbf{F}_{rep} + \mathbf{F}_{att}.$$

In the proposed framework, the SFM is extended to a multi-modal formulation. The model parameters are dynamically selected based on the GNN output: θ_t = θ^(k_t). This design enables the simulation to adapt its behavior according to the inferred interaction mode, effectively forming a hybrid dynamical system with discrete mode switching and continuous state evolution. A key feature of the platform is the coupling between the GNN and the simulation engine. The GNN output serves as a control signal that determines the parameter set used in the SFM. This interaction can be expressed as:

$$\mathcal{G}_t \mathrm{GNN} \hat{k}_t \theta^{(\hat{k}_t)} \mathrm{SFM} \mathbf{x}_{t+1}.$$

This modular interface ensures that the learning component influences the simulation only through interpretable parameters, preserving physical consistency. To calibrate model parameters, the platform incorporates a numerical optimization layer based on the Discrete Adjoint-State Method. This approach enables efficient gradient computation for time-dependent systems by solving an adjoint system backward in time. The objective function combines mode classification loss and trajectory reconstruction error:

$$\mathcal{L} = \mathcal{L}_{\text{mode}} + \lambda \mathcal{L}_{\text{traj}},$$

Where:

$$\mathcal{L}_{\text{traj}} = \sum_t \sum_i \left\| \mathbf{p}_{i,\text{sim}}^{(t)} - \mathbf{p}_{i,\text{obs}}^{(t)} \right\|_2^2.$$

The adjoint method enables efficient propagation of gradients through the simulation, facilitating parameter updates without requiring computationally expensive finite-difference approximations.

4. Simulation/simulator /simulation model

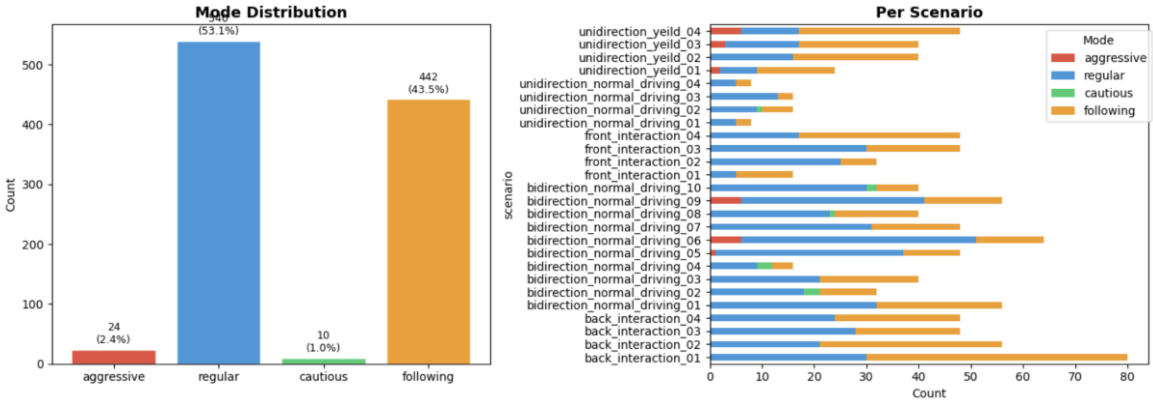
In the simulation section, what we implemented was a layered hybrid simulator. The bottom layer is GSFM forward simulator. It reads the pedestrian and vehicle trajectories by frame. In each time step, calculate the pedestrian interaction, pedestrian vehicle interaction and anisotropy, and then update the speed and position based on the current combined force. In the code, we write this process as a frame-by-frame forward simulation, so essentially it is a discrete time-stepping method. We choose this approach mainly because data like DUT-CITR itself are discrete trajectories recorded by frame. This approach is more direct in aligning with the original data and also more convenient for subsequent trajectory comparison. For our project, this method is more suitable than complex continuous-time solvers because it has a lower implementation cost, clearer debugging, and is easier to explain the source of dynamics at each step. On this basis, we extended the single-parameter GSFM to Multi-Modal GSFM. Now the simulator will receive the mode probabilities of each pedestrian, then calculate the forces using four sets of mode-specific parameter sets respectively, and finally weight these forces according to the probabilities. Obtain the total force of the agent at the current moment. And the current system supports four behavior modes: aggressive, regular, cautious and following. We chose this method because it can inject behavior uncertainty into the simulator in the form of soft. In this way, the system not only retains the interpretability of the force-based model, but also can represent the behavioral changes of the same pedestrian in different interaction contexts. Compared with using only one set of fixed parameters, this approach is closer to the real pedestrian-vehicle interaction.

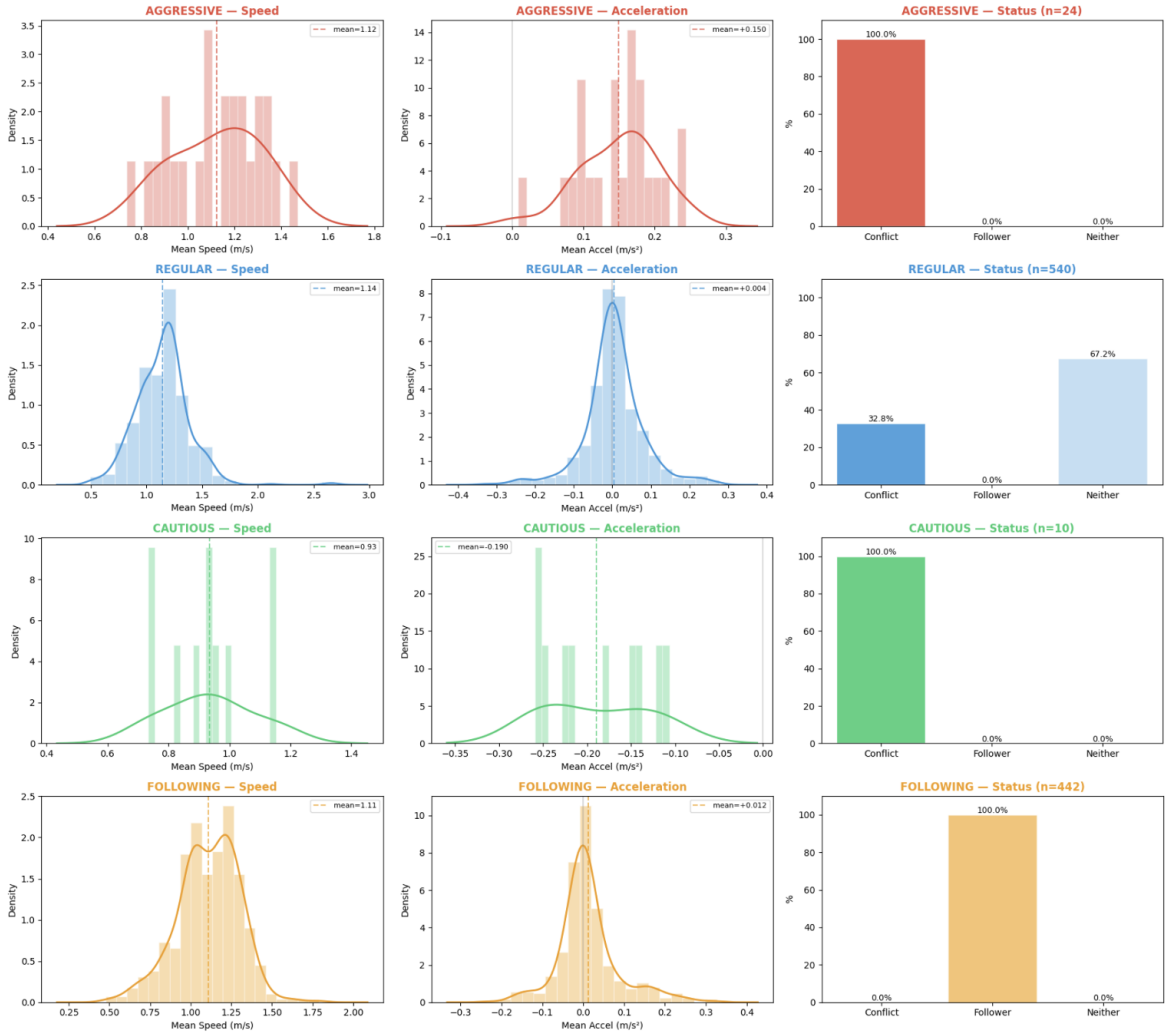
At the behavior recognition layer, we did not have the neural network directly predict future coordinates. First, we construct supervision from the TTC evolution and local motion information using pseudo-labeling, and then let the classifier output behavior mode probability. The current rule divides behaviors into four categories. Among them, aggressive and cautious both depend on TTC changes and the result of yielding, while following depends on crowd-level alignment. Feature construction is also divided into two parts. The time series features of the target pedestrian include x, y, vx, vy, speed, accel and heading_change. In the local graph structure, node features use x, y, vx, and vy, while edge features use distance, bearing, and relative_speed. What we really want the model to learn are interaction patterns and behavior modes. This design is more easily connected with the subsequent physics layer. In the

implementation of classifier, the current main line we use is GRU plus graph encoder. The GRU is responsible for encoding the short-term motion sequence of the target pedestrian, the graph encoder is responsible for encoding the current local interaction graph, and then four types of softmax probabilities are output through the fusion head. We chose this structure because it is quite compatible with the current task scale. The behavior judgment of pedestrian is inherently related to short-term speed changes and local neighborhood relationships. Therefore, using a lightweight temporal encoder combined with graph message passing can already express the main information. This structure also has a practical advantage, which is that it is very natural with the MM-GSFM interface at the back, and the output is directly mode probabilities. In contrast, although fully end-to-end trajectory prediction may be stronger, it will mix behavior recognition and motion generation together. It becomes very difficult to analyze separately why simulator generates a certain trajectory later on.

We have also considered other methods. We attempt to implement it using the Transformer. However, at the current stage, we still take the GRU based model as the main route. There are two reasons. First, the current data scale and the quality of pseudo-labels are still being improved, and the training stability and parameter tuning cost of the Transformer are higher. Secondly, our focus is to stably feed the behavior mode into the simulator. For the output analysis of the simulator, we have also implemented a complete numerical evaluation pipeline. Run_hybrid_pipelin.py first performs mode probability inference and then sends the results to MM-GSFM to generate simulated trajectories. After that, evaluate_hybrid.py will align the simulated trajectory with the observed trajectory. And calculate ADE, FDE, RMSE, speed error, path length error, minimum pedestrian-vehicle distance error And collision related consistency. We choose these indicators because merely looking at the trajectory position error is not sufficient. For pedestrian and vehicle simulation, whether the interaction is reasonable, whether the deviation of the minimum distance is too large, and whether the collision relationship is consistent are equally important. This set of indicators can more comprehensively reflect the numerical performance of the simulator.

5. Experimental results and validation

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AGGRESSIVE — Acceleration
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Mean Accel (m/s²)
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0.0%
%
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Follower
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Density
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Density
Mean Accel (m/s²)
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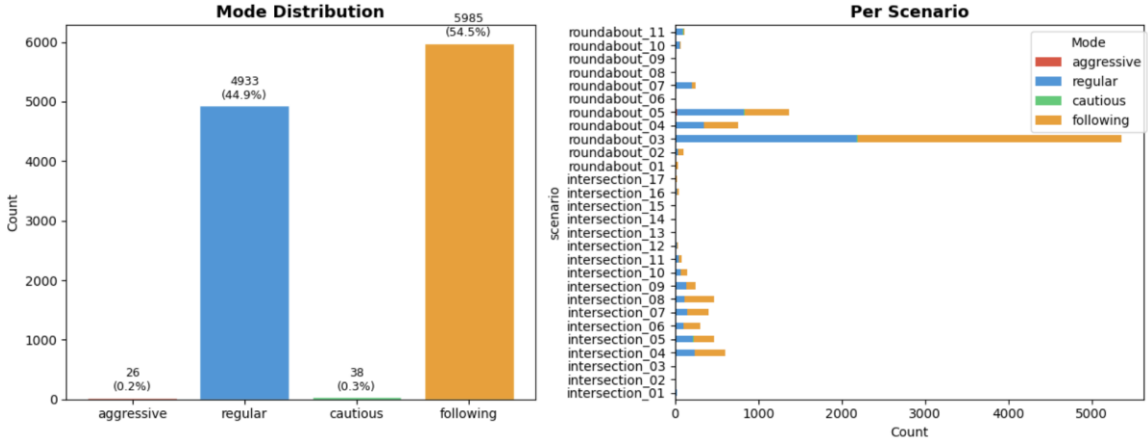
Follower
Neither


Mode Distribution
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4933
(44.9%)
38
(0.3%)
5985
(54.5%)
aggressive
regular
cautious
following
Per Scenario
scenario
Count
Mode
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regular
cautious
following
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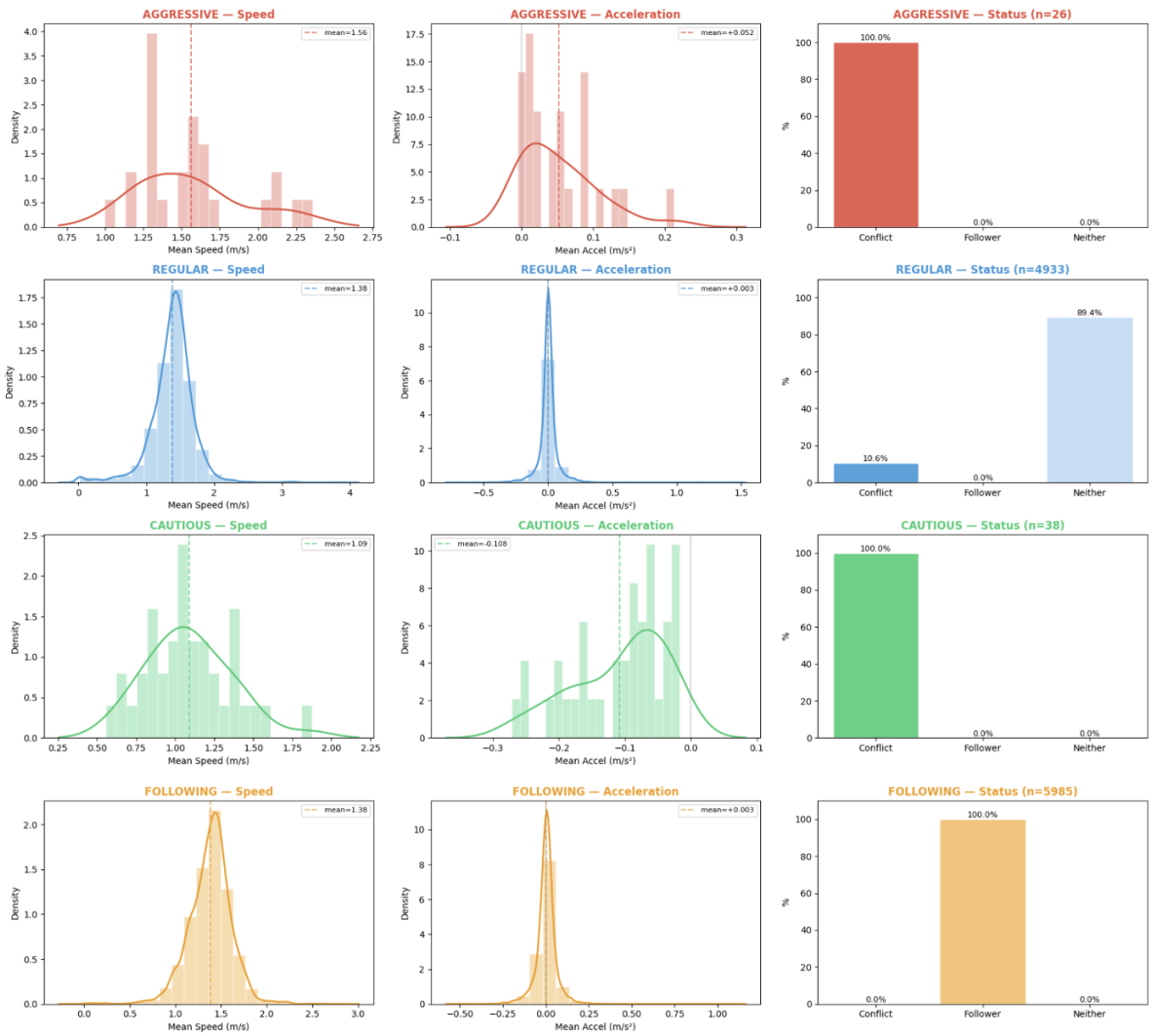

Figure 1. Behavior Mode Distribution

The proposed hybrid simulator, integrating a Graph Neural Network (GNN) for behavioral inference with a Social Force Model for motion dynamics, was evaluated using pedestrian–vehicle interaction trajectories from the DUT–CITR dataset. A total of eight pedestrians were analyzed over a shared temporal window of 421 frames (frames 311–731). For each pedestrian, simulated trajectories were compared against ground truth using position, velocity, and safety-related metrics.

Trajectory fidelity was evaluated using Average Displacement Error (ADE), Final Displacement Error (FDE), and Root Mean Square Error (RMSE). The results show a clear discrepancy between overall trajectory accuracy and endpoint accuracy.

The mean ADE across all pedestrians is:

$$\text{ADE} = 4.89 \pm 0.10\ m \quad (95\%\ CI),$$

while the mean FDE is significantly lower:

$$\text{FDE} = 0.016 \pm 0.019\ m \quad (95\%\ CI).$$

This indicates that although the simulator is highly accurate in predicting the final positions of pedestrians, it fails to reproduce the intermediate trajectory paths. Individual pedestrian results further confirm this pattern; for example, Pedestrian 1 exhibits an ADE of 5.09~m while achieving an FDE of only 0.005~m.

This large discrepancy between ADE and FDE suggests that the model is likely biased toward endpoint convergence, potentially due to implicit constraints or trajectory smoothing effects. As a result, the simulated trajectories deviate substantially from real-world paths despite ending at nearly identical positions.

Velocity-related metrics show moderate agreement with ground truth. The mean speed mean absolute error (MAE) across all pedestrians is approximately 0.71~m/s, with a corresponding RMSE of 0.80~m/s. These values indicate that the simulator captures the general scale of pedestrian motion but lacks precision in temporal dynamics.

Path length comparisons further reveal inconsistencies. While ground truth path lengths range from approximately 13~m to 15~m, simulated trajectories exhibit absolute errors up to 1.26~m. This suggests that the simulator does not accurately reproduce the curvature or geometric characteristics of real trajectories, leading to either over-simplified or distorted paths.

The GNN-based behavioral inference model predicts pedestrian modes with a strong bias toward the ``following" category. Specifically, 7 out of 8 pedestrians are classified as following, with mean probability values:

$$P(following) = 0.686, \quad P(regular) = 0.266, \quad P(cautious) = 0.031, \quad P(aggressive) = 0.017.$$

This distribution suggests that the model captures a dominant behavioral trend consistent with pedestrian responses in vehicle interaction scenarios. However, the lack of diversity in predicted modes indicates limited sensitivity to varying behavioral patterns, which may contribute to the observed trajectory inaccuracies.

Safety performance was evaluated using collision rate and minimum pedestrian–vehicle distance. In the ground truth data, no collisions were observed. In contrast, the simulator predicts collisions for 2 out of 8 pedestrians, corresponding to a collision rate of:

$$\text{Collision Rate} = 0.25 \pm 0.30 \quad (95\%\ CI).$$

The large confidence interval reflects the limited sample size but still indicates a substantial overestimation of collision risk. Additionally, the mean absolute error in minimum pedestrian–vehicle distance exceeds 1~m for several pedestrians, further highlighting discrepancies in interaction modeling.

These results suggest that the repulsive force or interaction modeling between pedestrians and vehicles is not adequately calibrated, leading to unrealistic proximity behaviors and unsafe simulated outcomes.

The evaluation reveals that the proposed hybrid simulator achieves strong performance in endpoint prediction but exhibits significant limitations in trajectory fidelity and safety realism. The consistently low FDE demonstrates that the model can successfully predict final pedestrian positions, which is valuable for certain applications such as destination estimation. However, the high ADE indicates that the underlying motion dynamics are not accurately captured.

The mismatch between simulated and real trajectories likely arises from a combination of factors, including simplified force modeling, insufficient temporal constraints, and limited behavioral diversity. Furthermore, the overestimation of collision rates suggests that the interaction forces governing pedestrian–vehicle avoidance are not properly tuned.

Another important limitation is the small sample size (N=8), which restricts the statistical robustness of the results. While confidence intervals for ADE are relatively tight, safety-related metrics exhibit high uncertainty. Future work should incorporate larger datasets and scenario diversity to improve both model calibration and statistical reliability.

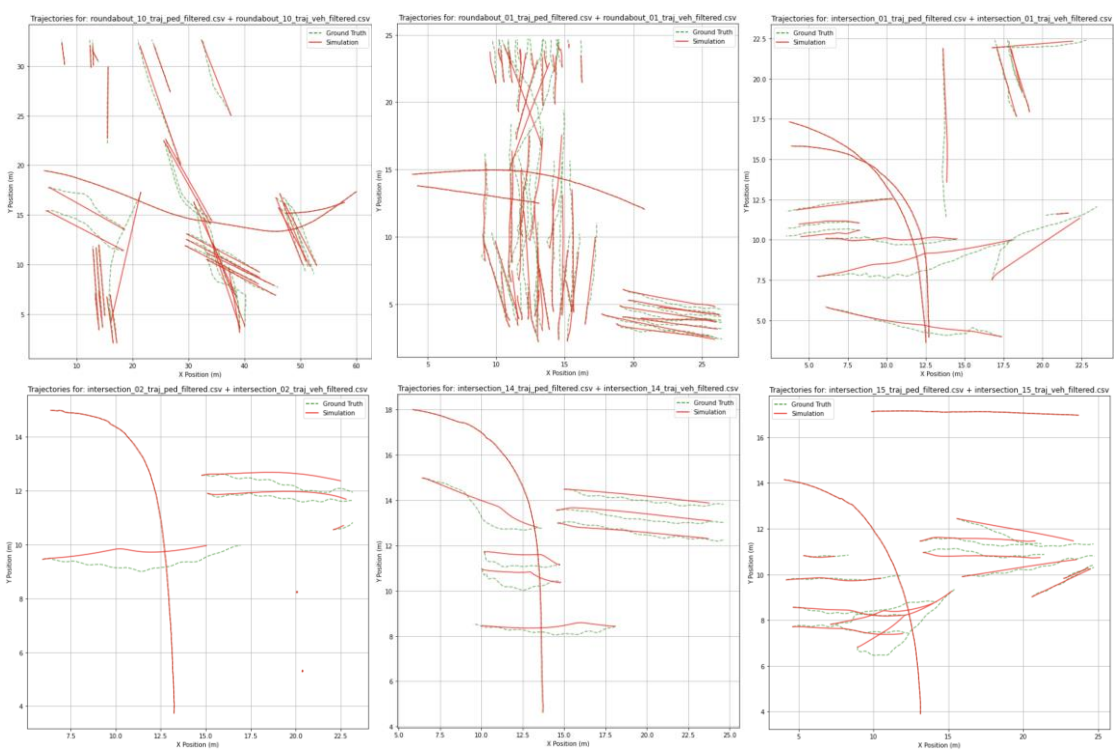

Figure 2. Trajectory Overlay (ground truth with simulation)

To evaluate the sensitivity of the simulator to model parameters, a multi-generation parameter search was conducted over key social force parameters, including pedestrian–pedestrian

interaction strength, pedestrian–vehicle interaction strength, desired speed, and anisotropy coefficients. Each parameter configuration was evaluated using the average position error metric.

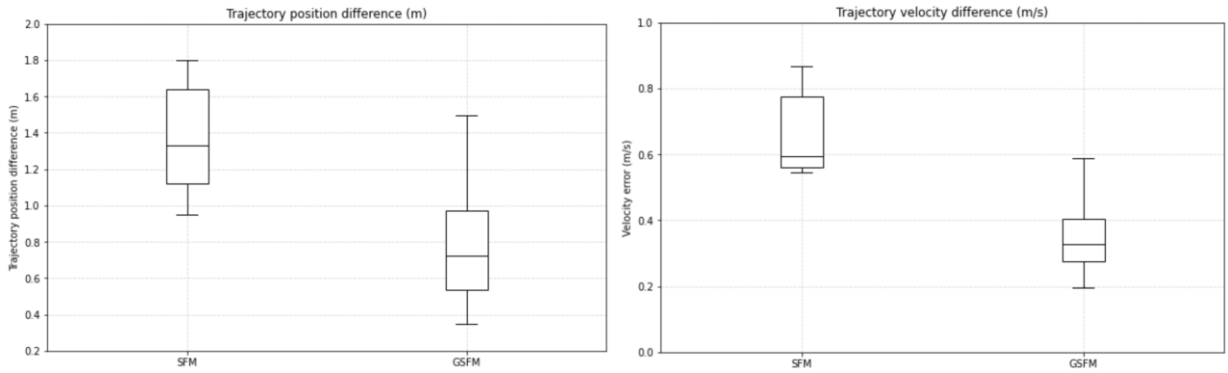

Figure 3. Trajectory position difference(left); Trajectory velocity difference(right)

Results show that the average position error varies significantly across parameter configurations, ranging from approximately 1.91~m in poorly tuned cases to as low as 0.63~m in optimized configurations. The best-performing parameter set achieved an average position error of 0.6278~m in the second generation, compared to typical values around 0.9–1.0~m.

These results demonstrate that the simulator is highly sensitive to parameter selection, and that appropriate calibration can substantially improve trajectory prediction accuracy. In particular, higher pedestrian–vehicle interaction strength and lower reaction time constants were frequently observed in high-performing configurations, suggesting that accurate modeling of fast and strong interaction responses is critical for realistic simulation.

However, the large variability across parameter sets also indicates that the model lacks inherent robustness and requires careful tuning to achieve reliable performance.

6. Discussion, conclusions, summary

The simulated pedestrian–vehicle interaction system is highly parameter-sensitive. Small changes in social-force parameters, especially pedestrian–vehicle interaction strength, reaction behavior, desired speed, and destination weighting, can strongly change the trajectory error. The optimization log shows average position error varying from about 0.63 m to 1.9 m, meaning calibration is essential rather than optional.

The system also shows a clear mismatch between endpoint accuracy and path realism. The simulator can reproduce final pedestrian positions very well, but the intermediate trajectories can deviate significantly. This suggests that the model may be too strongly constrained by destination or endpoint behavior, while not fully capturing the real-time decision process of pedestrians.

Safety behavior is another important finding. Ground truth showed no collisions, but the simulator predicted collisions for some pedestrians. This means the current pedestrian–vehicle interaction force is not yet realistic enough for safety-critical simulation.

Future work should first expand the validation dataset. The current result uses only eight pedestrians in one scenario, so the model should be tested across more DUT/CITR scenes with different vehicle speeds, pedestrian densities, crossing angles, and interaction types.

Second, the simulator should include a more systematic parameter calibration method, such as Bayesian optimization, genetic algorithms, or gradient-free search. Since the results show strong parameter sensitivity, calibrated parameters should be reported with confidence intervals instead of using only one fixed parameter set.

Third, the behavior model should be improved. The current GNN predicts mostly “following” behavior, so future work should balance the training data or redesign behavior labels to better distinguish cautious, aggressive, and yielding pedestrians.

Finally, the simulator should be validated not only using ADE/FDE but also using safety and interaction metrics such as minimum pedestrian–vehicle distance, time-to-collision, yielding rate, and collision consistency. This would make the simulator more useful for autonomous vehicles and traffic safety studies.

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Appendix: Project GitHub Link:

https://github.com/Austin-Jiang-xlk/CSE-6730-Project

Division of labor

Briefly explain how you divided the work. Who did what?

- Xisheng Zhang: Implement GNN model/config/training/pipeline
- Jiacheng Wang: GNN + Transformer + MM-GSFM platform
- Keller Smith: Writing checkpoint 2 report/ Evaluation
- Haotian Jiang: Multi-Modal GSFM Simulation

The division of labor for this project is arranged by system modules, so that each person can be responsible for a relatively complete work line. Xisheng Zhang is mainly responsible for GNN model, configuration, training pipeline and related code integration, while Jiacheng Wang is mainly responsible for the exploration of GNN and Transformer directions. And participated in the construction of the MM-GSFM platform. Keller Smith was mainly responsible for writing the checkpoint 2 report and designing and organizing the evaluation section. Haotian Jiang is

mainly responsible for the implementation of the Multi-Modal GSFM simulation and pseudo labeling modules. Overall, each member first advances their own module respectively, and then in the later stage, they focus on interface connection, experimental integration and result summary.